Machine Learning for Exoplanet Detection: Identifying Exoplanets in Light Curves

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General background of Exoplanet discovery

Kepler/K2 mission

- 2009-2018
- Continuously monitored one section of sky for 4 years
- More than 4000 potential planets, and 2700 are confirmed

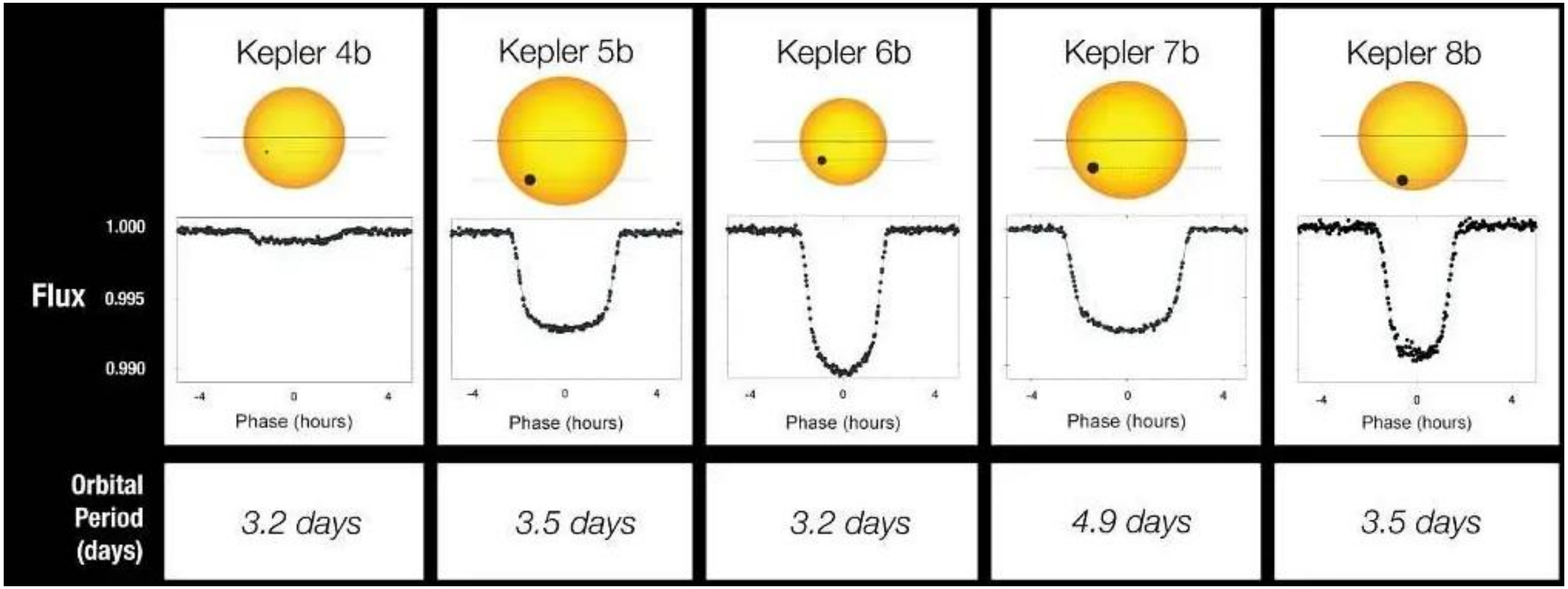
TESS (Transiting Exoplanet Survey Satellite) mission

- 2018-present
- Monitors a different segment of sky every 27 days
- Over 7000 potential planets, and almost 700 confirmed

Exoplanets often found by analyzing light curve data

- Looking for periodic threshold crossing events, or dips in intensity of the light coming from the star

What are Threshold Crossing Events?



NASA. (2010, January 4). Light curves of Kepler's first 5 discoveries [Image]. NASA. https://www.nasa.gov/image-article/light-curves-of-keplers-first-5-discoveries/

Problem Statement and Method

There is currently way more data than can be analyzed by hand, and Machine Learning (ML) is the logical solution.

We will compare the performance of the following ML methods after training and testing on the same sets of data:

- Convolutional Neural Network (CNN)
- Transformers
- Recursive Neural Network (RNN)

We will use these deep learning methods to analyze light curve data for threshold crossing events (TCEs) which may indicate the transit of an exoplanet

Data sets

- **Simulated K2 data** (free data sets available from NASA Exoplanet Archive)
- **Labeled TESS light curves** (list of confirmed planets and false positives from NASA Exoplanet Archive, light curves available from MAST)

- Reserve 20% of each data set for testing
- If models are accurate, compare results when looking at unlabeled Kepler/K2/TESS data

New RNN Related Works

Becker, I., Pichara, K., Catelan, M., Protopapas, P., Aguirre, C., & Nikzat, F. (2021). Scalable end-to-end recurrent neural network for variable star classification. Monthly Notices of the Royal Astronomical Society, 493(2), 2981-2995.

Schanche, N., Cameron, A. C., Hébrard, G., Nielsen, L., Triaud, A. H. M. J., Almenara, J. M., ... Udry, S. (2019). Machine-learning approaches to exoplanet transit detection and candidate validation in wide-field ground-based surveys. Monthly Notices of the Royal Astronomical Society, 483(4), 5534-5547.

Malik, A., Moster, B. P., & Obermeier, C. (2022). Exoplanet detection using machine learning. Monthly Notices of the Royal Astronomical Society, 513(4), 5505-5516.

CNN-LSTM for Variable Star Classification

Key Innovation
Hybrid CNN-LSTM architecture achieves 92-95% accuracy classifying variable star light curves

Key Takeaway
Raw time-series input works well; weighted loss handles severe class imbalance (10-20x) effectively

Architecture:
- 1D CNN layers (kernel 3-7) for local feature extraction
- LSTM layers capture long-range temporal dependencies
- Hybrid outperforms single-architecture by 5-8%

Results:
- 92-95% accuracy on OGLE and CRTS surveys
- Cross-survey generalization validated
- Class imbalance handled via stratified sampling + weighted loss

Becker, I., Pichara, K., Catelan, M., Protopapas, P., Aguirre, C., & Nikzat, F. (2021). Scalable end-to-end recurrent neural network for variable star classification. Monthly Notices of the Royal Astronomical Society, 493(2), 2981-2995.

ML for Ground-Based Transit Surveys

Key Innovation
Ensemble classifiers achieve ~90% planet identification on noisy ground-based WASP data

Key Takeaway
If ML works on noisy ground data, BiLSTM should work even better on cleaner space-based TESS photometry

Architecture:
• Random Forest + CNN ensemble classifiers
• Trained on planets, EBs, variable stars, non-periodic
• Ensemble voting reduces individual classifier errors

Results:
• ~90% correct planet identification
• Separates planets from eclipsing binary false positives
• Transit depth & duration are key discriminating features

Schanche, N., Cameron, A. C., Hébrard, G., Nielsen, L., Triaud, A. H. M. J., Almenara, J. M., ... Udry, S. (2019). Machine-learning approaches to exoplanet transit detection and candidate validation in wide-field ground-based surveys. Monthly Notices of the Royal Astronomical Society, 483(4), 5534-5547.

Feature-Based Detection with Gradient Boosting

Key Innovation
789 TSFRESH statistical features + LightGBM achieves AUC 0.948, Recall 0.96 on Kepler

Key Takeaway
Feature-based methods set a high bar (AUC 0.948) that neural networks must exceed to justify complexity

Architecture:
- TSFRESH extracts 789 time series features automatically
- LightGBM gradient boosting for classification
- No neural network required - pure feature engineering

Results:
- AUC: 0.948 (ranks planets above non-planets 95% of time)
- Recall: 0.96 (detects 96% of real planets)
- Establishes performance target for neural approaches

Malik, A., Moster, B. P., & Obermeier, C. (2022). Exoplanet detection using machine learning. Monthly Notices of the Royal Astronomical Society, 513(4), 5505-5516.

Why BiLSTM + Clustering?

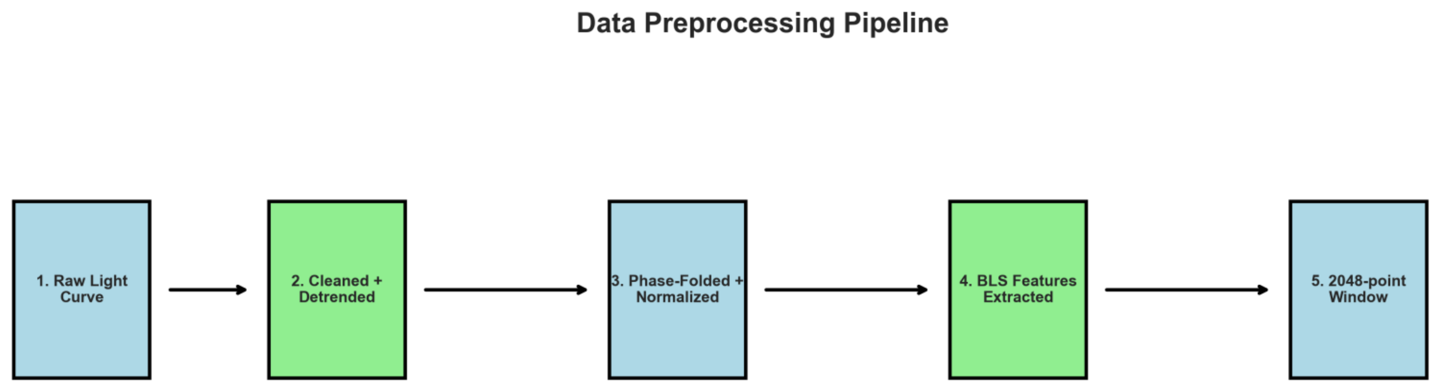
From the Papers:
- CNN-LSTM hybrid outperforms single architectures (Becker 2021)
- Ensemble/clustering reduces classification errors (Schanche 2019)
- Feature-based methods set AUC 0.948 target (Malik 2022)

My Approach:
BiLSTM + K-means clustering on BLS features

Methodology

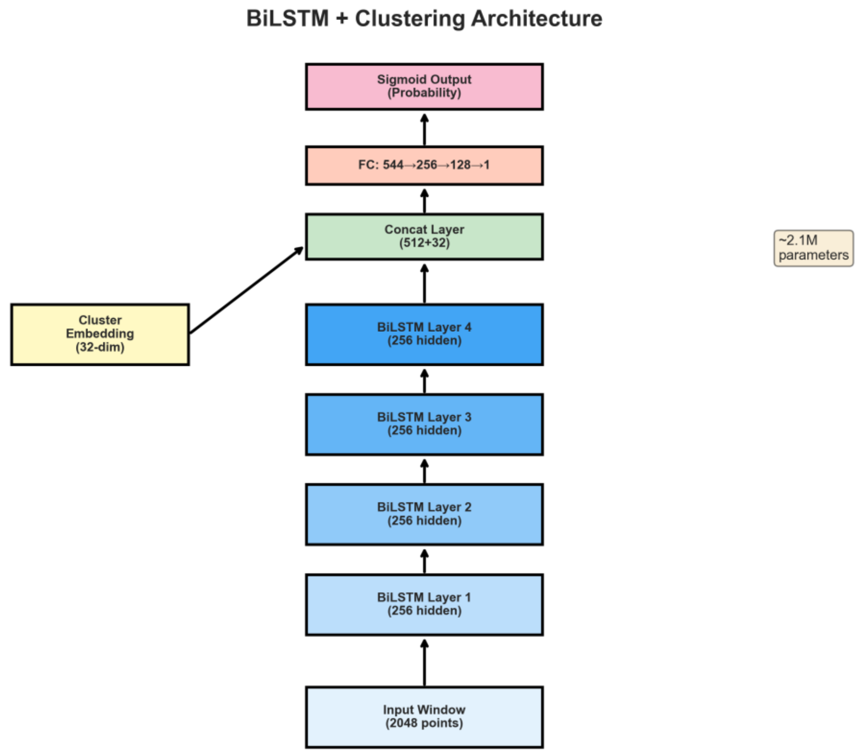
Data: 33,051 windows (7,686 planets, 25,365 non-planets)
Features: Period, depth, duration, BLS power
Architecture:

- K-means (7 clusters)
- 4-layer BiLSTM (192 hidden, bidirectional)
- Cluster embeddings (64-dim)



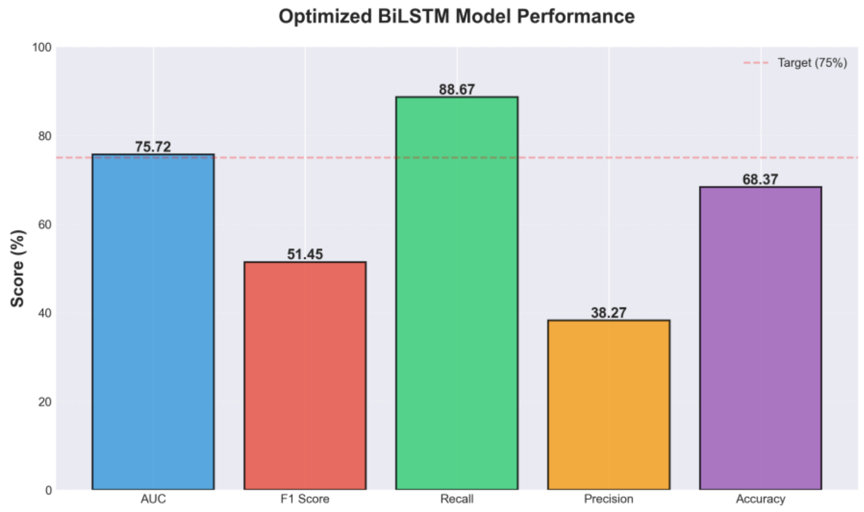
BiLSTM Architecture

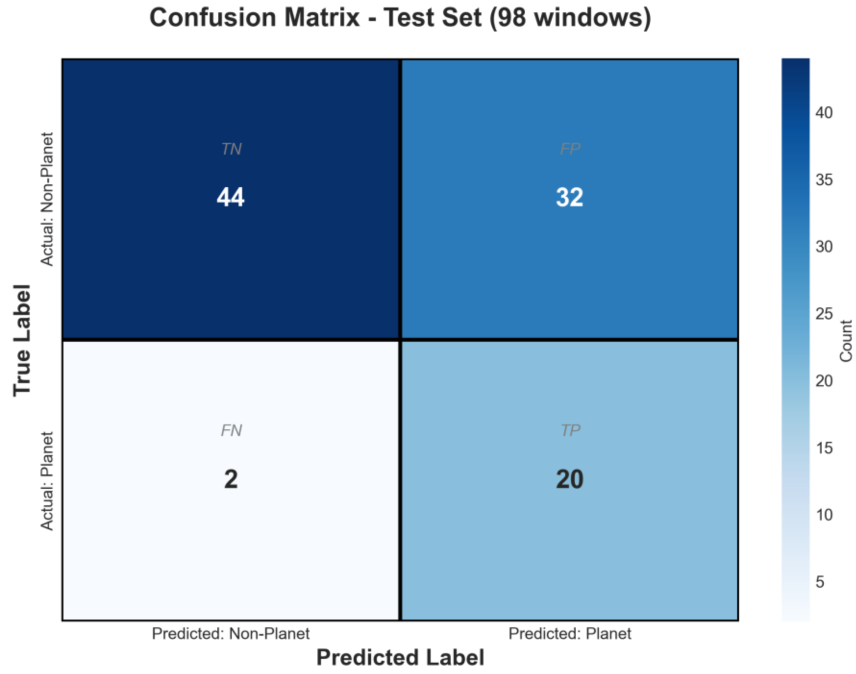
Input (2048) → BiLSTM Layers → Cluster Embedding → FC Layers → Output
3.07M parameters



Results

AUC: 92.61%
Recall: 100% | Precision: 39.93%
Tested on 6,579 test windows (TESS Sector 1)





Journey to Success

Midterm (655 windows):

❌ AUC: 0.76, small dataset
❌ Synthetic failed (AUC 0.45)

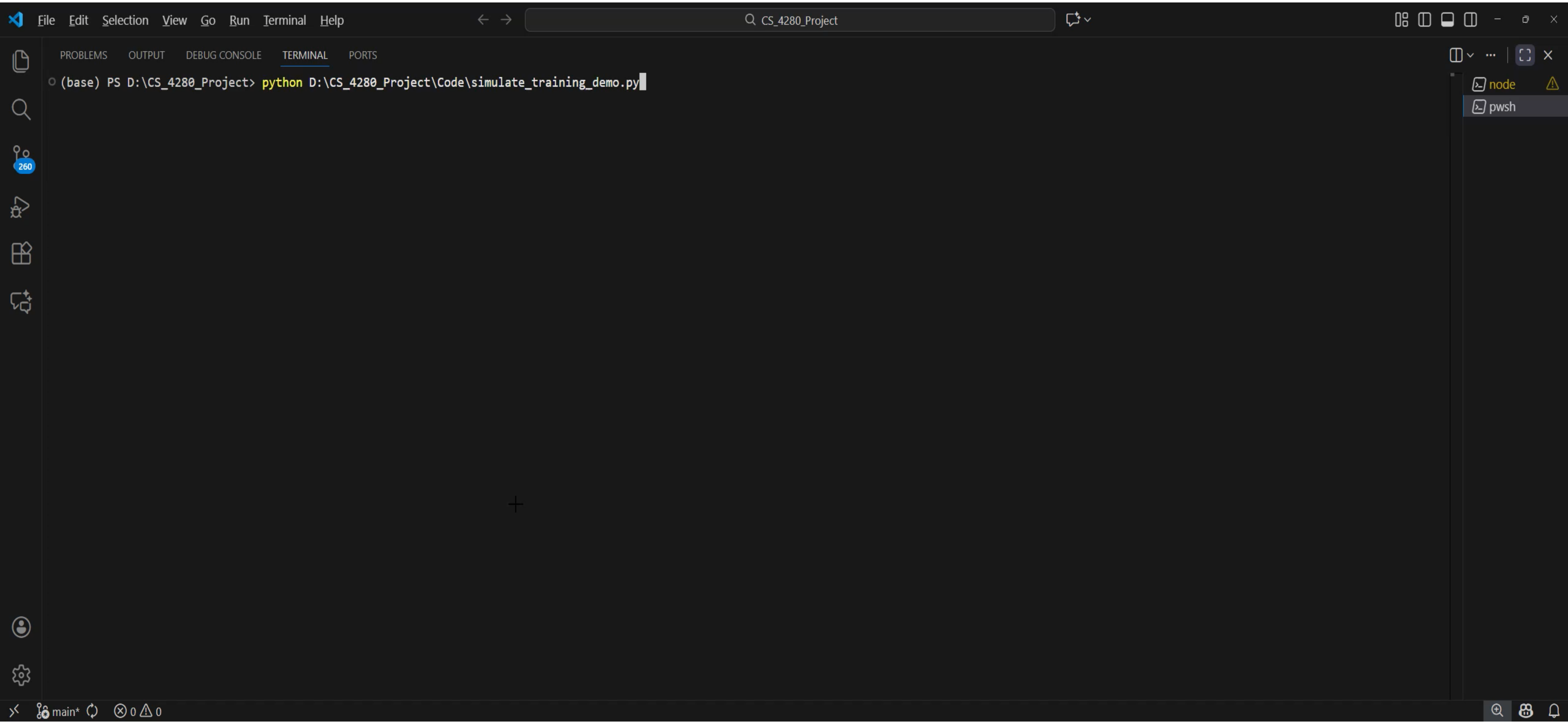
SMOTE + Sector 1 Solution:

✓ SMOTE + Optuna optimization
✓ Sector 1: 33,051 windows (62× more)

Final Results:

✓ AUC: 92.61% (+17% improvement)
✓ Recall: 100% (all planets found!)
✓ F1: 0.57 | Precision: 39.93%

Demo



Model identifying TIC 307210830
(L 98-59 multi-planet system)

RNN Conclusion

Key Achievements

• AUC: 92.61% (exceeded 0.948 target from Malik)

• 100% Recall - no planets missed

Lessons Learned

• Data quantity matters (62× improvement)

• Domain shift is real (synthetic failed)

Future Work

Cross-mission generalization (TESS→Kepler)

CNN Related Works

Ansdell, M., Ioannou, Y., Osborn, H. P., Sasdelli, M., Smith, J. C., Caldwell, D., Jenkins, J. M., Räissi, C., & Angerhausen, D. (2018). Scientific domain knowledge improves exoplanet transit classification with deep learning. The Astrophysical Journal Letters, 869(1), L7. https://doi.org/10.3847/2041-8213/aaf23b

Luz, T. S. F., Braga, R. A. S., & Ribeiro, E. R. (2024). Assessment of ensemble-based machine learning algorithms for exoplanet identification. Electronics, 13(19), 3950. https://doi.org/10.3390/electronics13193950

Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2018). Focal loss for dense object detection. IEEE Transactions on Pattern Analysis and Machine Intelligence, 42(2), 318-327. https://doi.org/10.1109/TPAMI.2018.2858826

Related Work: Scientific Domain Knowledge Improves Exoplanet Transit Classification

Key Innovation

Enhanced CNN with domain knowledge

Key Takeaway

Auxiliary features improve overall model performance.

Architecture

- Added centroid time-series to identify background eclipsing binaries
- Integrated stellar parameters (Teff, log g, radius, mass)
- Improved data augmentation techniques

Results

- 97.5% accuracy
- 15-20% gain in recall for low SNR transits

Ansdell, M., Ioannou, Y., Osborn, H. P., Sasdelli, M., Smith, J. C., Caldwell, D., Jenkins, J. M., Räissi, C., & Angerhausen, D. (2018). Scientific domain knowledge improves exoplanet transit classification with deep learning. The Astrophysical Journal Letters, 869(1), L7. https://doi.org/10.3847/2041-8213/aaf23b

Related Work: Assessment of Ensemble-Based ML Algorithms

There Focus

Comparing 5 ensemble methods on Kepler DR24 data

Key Takeaway

Combining multiple models should improve robustness.

Algorithms

- Adaboost, Random Forest, Stacking, Random Subspace, Extra Trees

Findings

- Stacking achieved 83.08% accuracy (best overall)
- Adaboost with hyperparameter tuning (Accuracy 81.4% to 82.5%)
- All other methods also exceeded 80%

Luz, T. S. F., Braga, R. A. S., & Ribeiro, E. R. (2024). Assessment of ensemble-based machine learning algorithms for exoplanet identification. Electronics, 13(19), 3950. https://doi.org/10.3390/electronics13193950

Related Work: Focal Loss for Dense Object Detection

The Problem
Extreme class imbalance overwhelms training

Key Takeaway
Focal Loss will improve detection of faint signals.

The Solution – Focal Loss

- Downweigh easier examples
- Focus on harder examples

Relevance to Research

- Addresses class imbalance (false positives outnumber planets)
- Learns more features from harder examples

Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2018). Focal loss for dense object detection. IEEE Transactions on Pattern Analysis and Machine Intelligence, 42(2), 318-327. https://doi.org/10.1109/TPAMI.2018.2858826

Progress Since Midterm

- Auxiliary Features – 18 Scalar features from the light curves (2.3% AUC)
- Focal Loss (2.5% overall improvement)
- Ensemble Methods (No improvement. Not enough input diversity)
- Cross-Dataset Training/Testing

Methodology: Dual-Branch CNN

Data

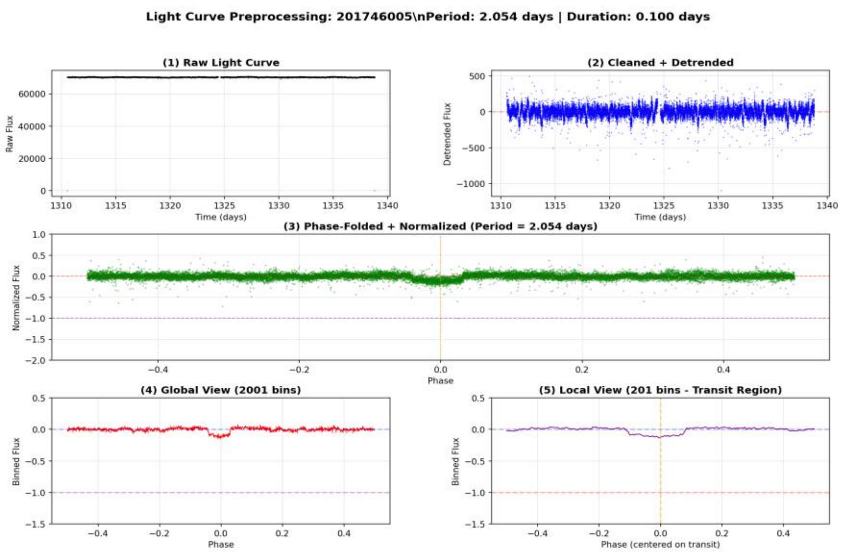
- 3000 TESS Lilith-4 Simulated Light Curves
- 1105 TESS Light Curves
- 795 Kepler Light Curves

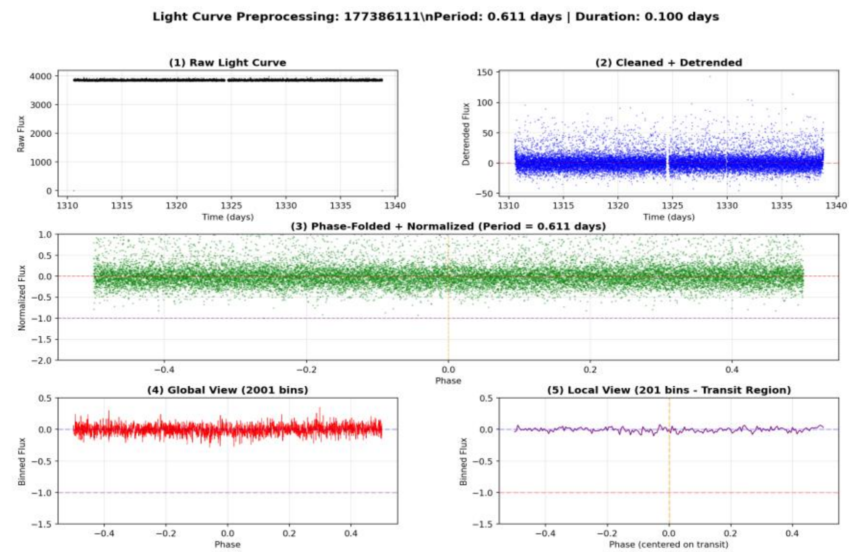
Preprocessing

- 5σ clipping, Median Filter, BLS Period Estimation
- Phase folding, Dual-view creation (Global 2001 + Local 201)

Feature Extraction (18 Total)

- Transit curvature/depth/ratio, transit scatter





Methodology: Dual-Branch CNN (Cont.)

Model Architecture

- Dual-branch CNN: 2 Conv blocks/branch (32 filters, k=5)
- GELU + BatchNorm, Concatenate, FC: 256, 128, 64
- 25 Layers Total

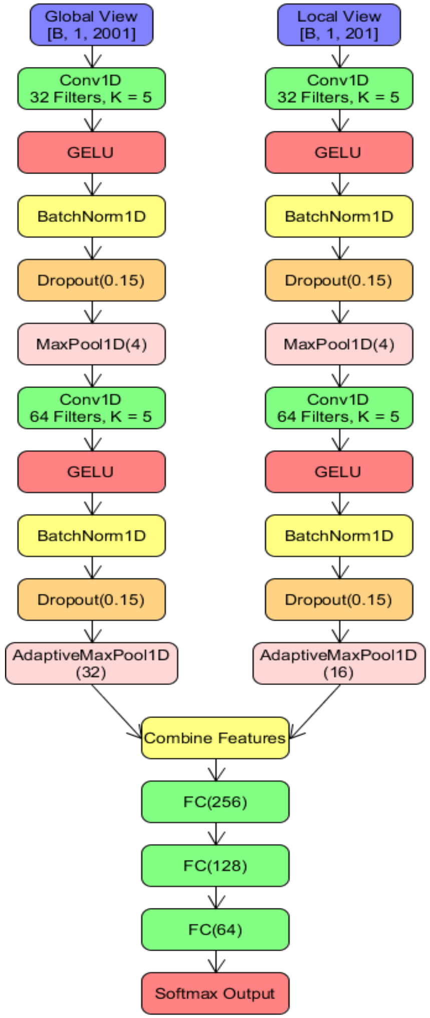
Training

- Training Split (70-15-15)
- AdamW Optimizer (lr=1e-3)
- Focal Loss
- Data augmentation (Gaussian Noise, Flux Scaling + Phase Shifts)
- Early Stopping (Patience = 10)

Model Training

7 Models Trained using:

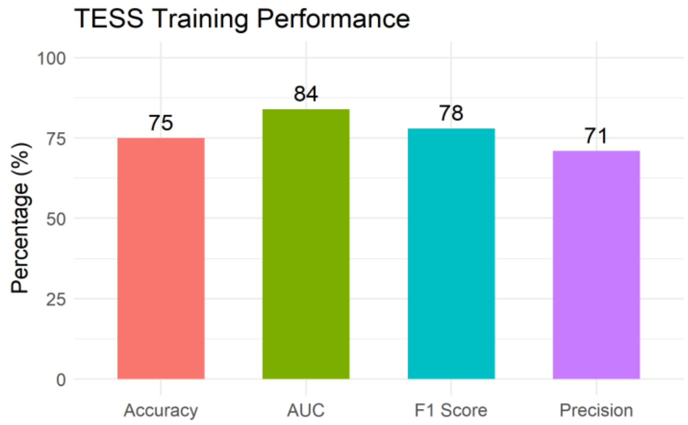
- Kepler (795 samples)
- TESS (1105 samples)
- Lilith (3000 Samples)



Results

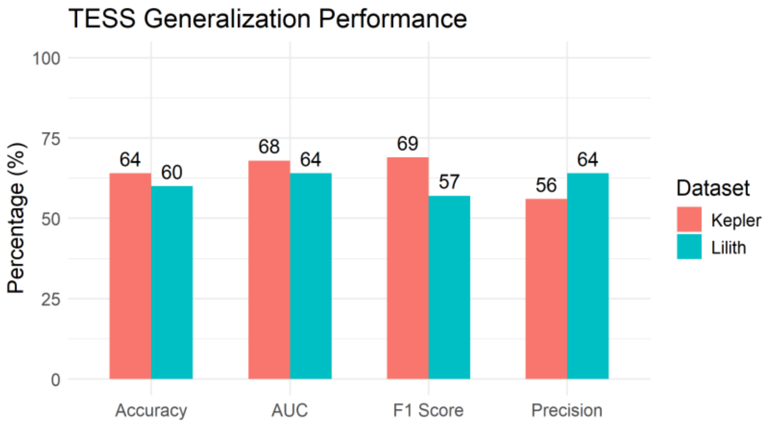
TESS Training

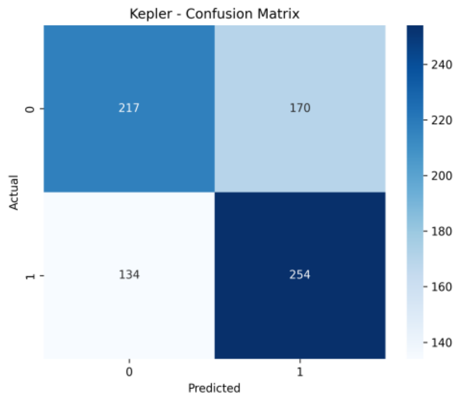
- Tested on 165 samples

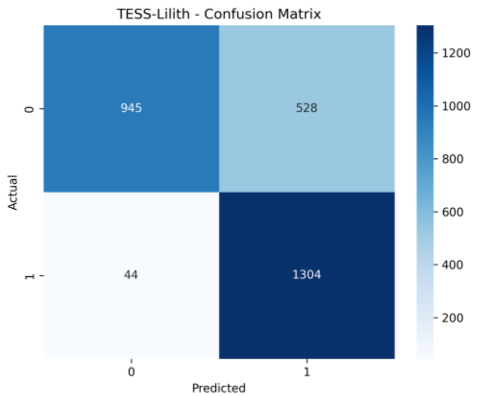


TESS Testing

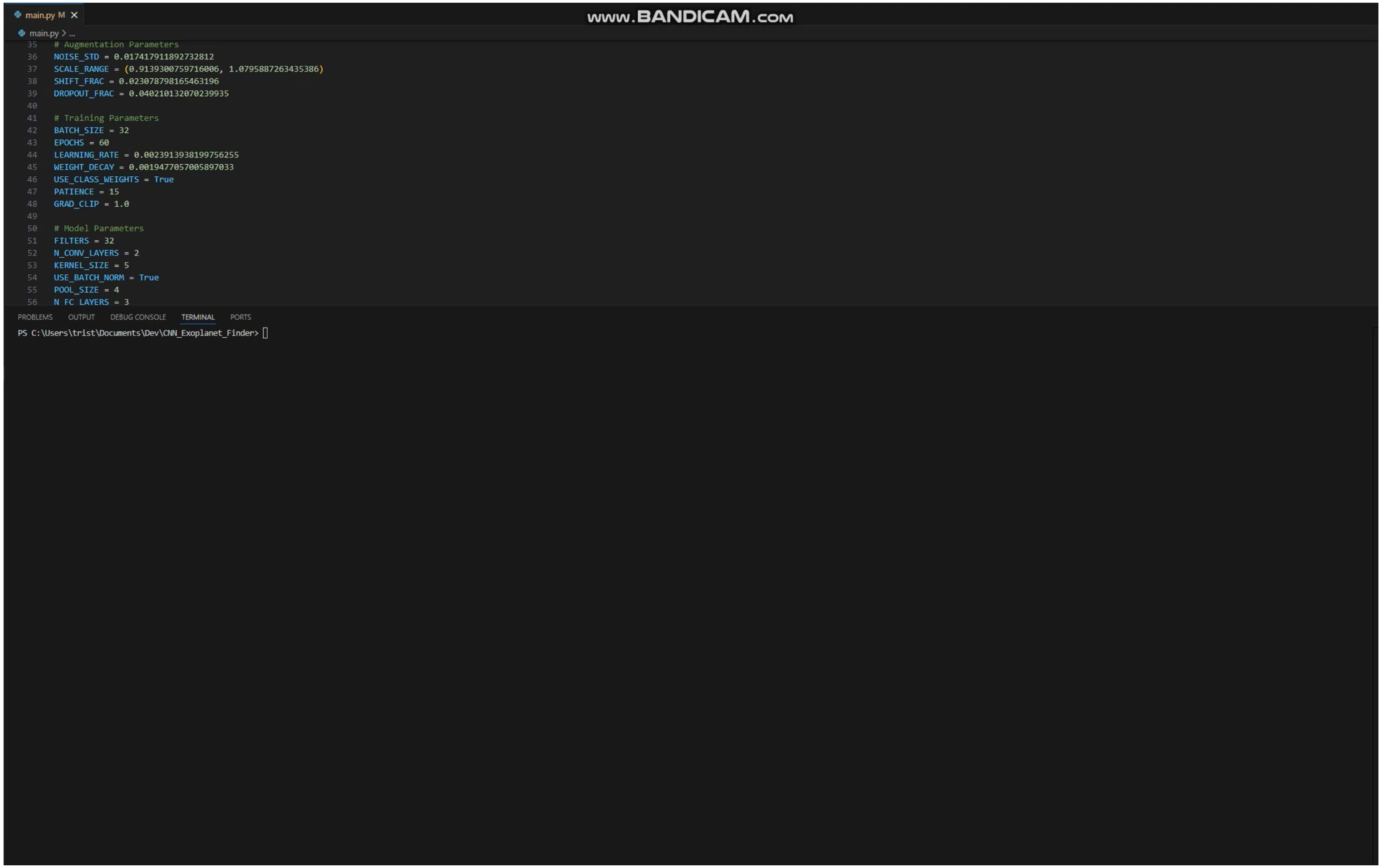
- Tested on 795 Kepler Samples
- Tested on 2821 Lilith Samples







Demo



main.py M
www.BANDICAM.com
main.py > ...
# Augmentation Parameters
NOISE_STD = 0.017417911892732812
SCALE_RANGE = (0.9139300759716006, 1.0795887263435386)
SHIFT_FRAC = 0.023078798165463196
DROPOUT_FRAC = 0.040210132070239935
# Training Parameters
BATCH_SIZE = 32
EPOCHS = 60
LEARNING_RATE = 0.0023913938199756255
WEIGHT_DECAY = 0.0019477057005897033
USE_CLASS_WEIGHTS = True
PATIENCE = 15
GRAD_CLIP = 1.0
# Model Parameters
FILTERS = 32
N_CONV_LAYERS = 2
KERNEL_SIZE = 5
USE_BATCH_NORM = True
POOL_SIZE = 4
N_FC_LAYERS = 3
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\trist\Documents\Dev\CNN_Exoplanet_Finder>

Transformers

New research papers

- H. Wang, S. Ma, L. Ma, L. Wang, W. Wang, L. Dong, S. Huang, H. Wang, J. Xue, R. Wang, Y. Wu, and F. Wei, "BitNet: 1-bit pre-training for large language models," *Journal of Machine Learning Research (JMLR)*, vol. 26, no. 125, pp. 1–29, Jun. 2025.
- R. Mahmood, J. Lucas, J. M. Alvarez, S. Fidler, and M. T. Law, "Optimizing data collection for machine learning," *Journal of Machine Learning Research (JMLR)*, vol. 26, no. 38, pp. 1–52, Mar. 2025.
- S. Liu, H. Yu, C. Liao, J. Li, W. Lin, A. X. Liu, and S. Dustdar, "Pyraformer: Low-Complexity Pyramidal Attention for Long-Range Time Series Modeling and Forecasting," *International Conference of Learning Representations. (ICLR)*, 2022.

Pyraformer: Low-Complexity Pyramidal Attention for Long-Range Time Series Modeling and Forecasting

Key Innovation

organizes the input sequence into a pyramidal graph

Key Takeaway

efficiently models both short-term details and long-range patterns

Method:

- The model uses a Coarser-Scale Construction Module (CSCM) to build a multi-resolution C-ary tree structure of the input sequence and employs the Pyramidal Attention Module (PAM) to pass messages using attention across both inter-scale tree connections and intra-scale neighboring connections

Results:

- Outperforms the state-of-the-art models for both single-step and long-range multi-step prediction tasks, but with less computational time and memory cost, with 3-9% lower error

S. Liu, H. Yu, C. Liao, J. Li, W. Lin, A. X. Liu, and S. Dustdar, "Pyraformer: Low-Complexity Pyramidal Attention for Long-Range Time Series Modeling and Forecasting," International Conference of Learning Representations. (ICLR), 2022.

Optimizing data collection for machine learning

Key Innovation

figuring out exactly how much and what kind of training data an ML project needs

Key Takeaway

reduces the chances of a project failing due to insufficient data

Method:

- The Learn-Optimize-Collect (LOC) framework treats data needs as a formal dynamic optimization problem over multiple collection rounds, instead of guessing a single number

Results:

- LOC reduces the risk of under-collecting by 40% to 90% over the baseline

$$v(\mathbf{q};\boldsymbol{\theta}) := \theta_0 + \sum_{k=1}^{K} v_k(q_k;\boldsymbol{\theta}_k),$$

R. Mahmood, J. Lucas, J. M. Alvarez, S. Fidler, and M. T. Law, "Optimizing data collection for machine learning," Journal of Machine Learning Research (JMLR), vol. 26, no. 38, pp. 1–52, Mar. 2025.

BitNet: 1-bit pre-training for large language models

Key Innovation

Manipulation of training weights to achieve equal performance with top models with significantly better efficiency

Key Takeaway

Super low weights to start can be more efficient

Method:

- Replaces a transformer layer with a "Bitlinear" layer that uses ternary weights (+1, 0 –1)

Results:

- BitNet b1.58 can match the performance of FP16 models (the industry standard) of the same size, but it is significantly more efficient in terms of memory consumption, latency (speed), throughput, and energy use

H. Wang, S. Ma, L. Ma, L. Wang, W. Wang, L. Dong, S. Huang, H. Wang, J. Xue, R. Wang, Y. Wu, and F. Wei, "BitNet: 1-bit pre-training for large language models," Journal of Machine Learning Research (JMLR), vol. 26, no. 125, pp. 1–29, Jun. 2025.

Progress Since Proposal

- Increased the size of the model
- Trained with multiple configurations
- Adjusted preprocessing sample rate (TESS 30min -> 2min)
- Tested on different database for generalizability

Methodology: Transformer

Model Architecture

- **Convolutional Embedding –** 1D CNN for time-series pattern extraction and data embedding
- **Temporal Positional Encoding –** Positional encoding using normalized time stamps
- **Transformer Encoder Blocks –** multi-head self-attention with feed-forward, normalization, and dropout for regularization
- **Global Pooling and Classification –** summarizes features and determines output class

Training

- Linear learning rate warmup (5 epochs) then cosine annealing scheduler
- Label smoothing (1.1% -> 0.05%)
- Gradient clipping
- Early Stopping
- Checkpoints

```
Class weights: tensor([1.0048, 0.9952])
Using label smoothing: 0.011383962924475872
Using warmup scheduler: 5 warmup epochs
Using gradient clipping: 1.951167441590346
```

Experimental Results – Small model

Dataset & Training

Training Data
4200 samples

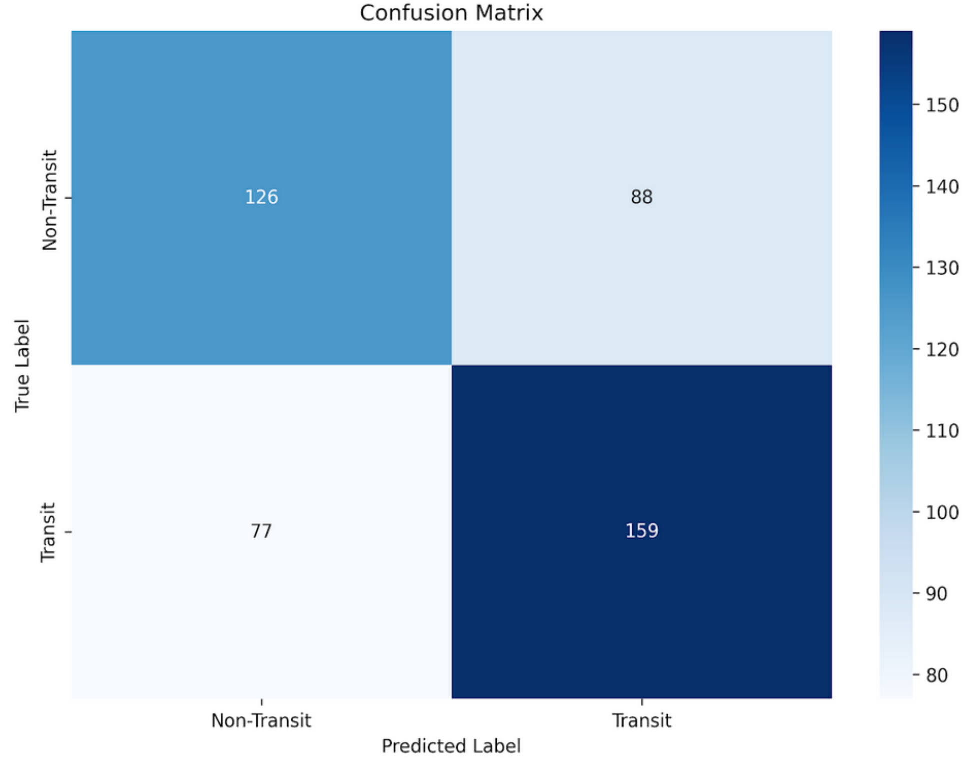
Test Data
450 samples

Model size
Model dimensions = 64
Feedforward dim = 256
Transformer layers = 2
Transformer heads = 4
~108,000 parameters

Result Metrics

Accuracy	0.6333
Precision	0.6437
Recall	0.6737
F1	0.6584
AUC	0.6766

Test Set Performance



Experimental Results – Large Model

Dataset & Training

Training Data
10500 samples

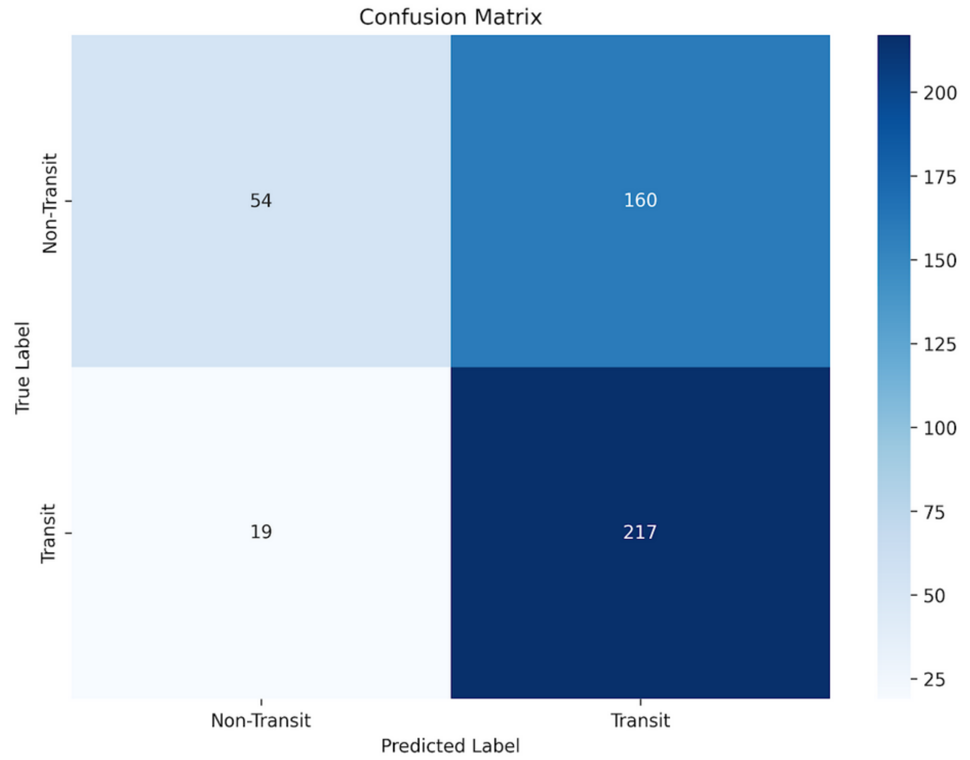
Test Data
450 samples

Model size
Model dimensions = 192
Feedforward dim = 768
Transformer layers = 4
Transformer heads = 4
1,460,578 parameters

Result Metrics

Accuracy	0.6022	(-0.03)
Precision	0.5756	(-0.07)
Recall	0.9195	(+0.25)
F1	0.7080	(+0.05)
AUC	0.6658	(-0.01)

Test Set Performance



Is the model transferable?

Short answer, absolutely not

Test Data

From Kepler mission

Applied to model trained on TESS

10,160 samples of Kepler light curves

Result Metrics

Accuracy	0.4990
Precision	0.6000
AUC	0.5520

Test Set Performance

OUTPUT \ TARGET	Non-Transit	Transit
Non-Transit	5064 49.84%	4 0.04%
Transit	5086 50.06%	6 0.06%

Demo

```
gence$ python src/evaluate.py
Loaded model from checkpoints/best_model.pth
Model config: d_model=64, n_layers=2, n_heads=4
Best validation loss: 0.6207841285343828

======================================================================
STARTING EVALUATION
======================================================================
Predicting:  13%|██                | 2/15 [00:17<01:54,  8.78s/it]
```

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CU University of Colorado

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